Advanced Computer Vision Systems

Modern visual representations, dense prediction, foundation models, and the reliability constraints of real-world perception.

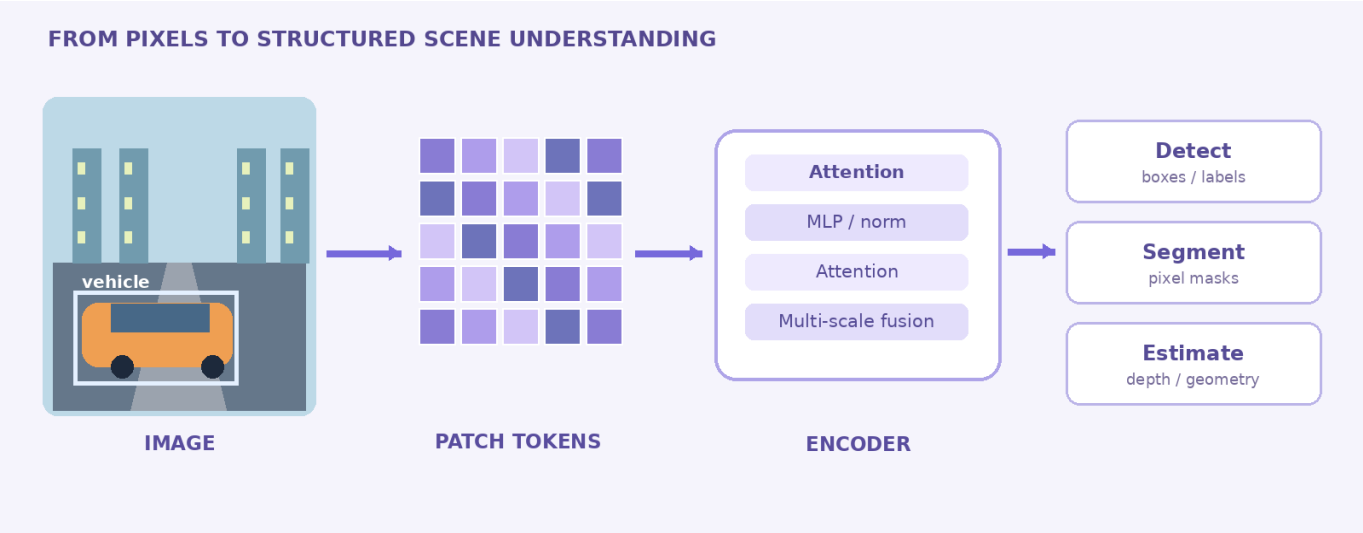


Figure 1. A shared visual encoder can support distinct prediction heads for object detection, segmentation, and geometry.

1 Visual representation learning

From convolution to tokens. CNNs encode translation-aware local features. Vision Transformers (ViTs) split images into patches, project them to token embeddings, and mix spatial information with attention. Global full attention scales quadratically with token count.

Self-supervised pretraining. Systems such as DINOv2 learn transferable features from images without requiring downstream class labels for every pretraining sample. The representation can then feed small task-specific classifiers, retrieval systems or dense predictors.

2 Perception beyond labels

Detection vs segmentation. A detector predicts boxes and categories; semantic segmentation predicts per-pixel classes; instance segmentation separates individual objects; panoptic segmentation combines "things" and "stuff" in a coherent map.

Promptable video masks. SAM 2 combines image segmentation with a streaming memory for tracking selected objects across frames. Production stacks may add geometry estimation, temporal filtering, tracking association, and camera calibration.

Attention scaling For H × W pixels and P × P patches, $N = HW / P^2$ tokens; dense self-attention uses $O(N^2)$ pairwise interactions per layer.

Metrics, complexity, robustness

TRADE-OFF MATRIX

Task	Typical output	Useful metric	Failure to inspect
Classification	Image-level category	Top-1 accuracy / AUROC	Class imbalance, shortcut cues
Object detection	Boxes + class scores	AP across IoU thresholds	Tiny / occluded objects
Segmentation	Dense pixel masks	mIoU / Dice score	Boundaries, rare classes
Depth estimation	Per-pixel distance	AbsRel / RMSE	Scale ambiguity, reflective surfaces
Tracking	IDs across frames	HOTA / IDF1	Identity switches, occlusion
Calibration	Confidence estimates	ECE / reliability plots	Overconfident wrong predictions

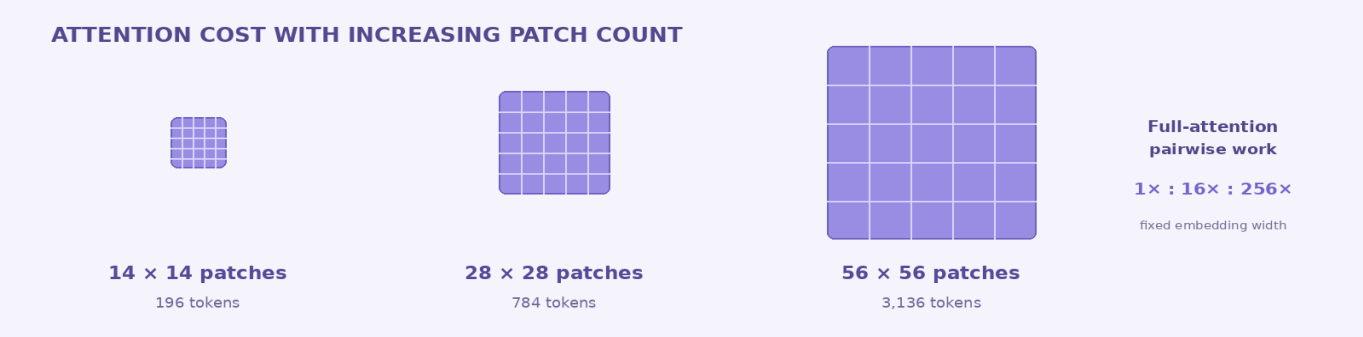


Figure 2. Doubling spatial patch resolution in each axis multiplies token count by 4 and pairwise work by 16.

Numerical design choices

A 224 × 224 RGB input with 16 × 16 patches gives **14 × 14 = 196 tokens** (excluding optional special tokens). Moving to 8 × 8 patches yields 784 tokens and approximately 16× more pairwise attention interactions.

Use hierarchical backbones, windowed attention, sparse attention, downsampling or patch selection where dense high-resolution attention is too costly. Verify actual memory and throughput rather than relying only on FLOPs.

Real-world evaluation protocol

Split data by **scene, site, time and identity**, not just random frames, to prevent near-duplicate leakage. Evaluate low light, motion blur, domain shift, weather, occlusions and sensor failures; track subgroup performance.

For high-stakes applications, verify calibration, test unknown-object behavior, monitor input drift, and apply human escalation or redundant sensing. An impressive benchmark score is not equivalent to field reliability.

PRIMARY SOURCES

ViT (ICLR 2021) DINOv2 (Meta) SAM 2 (Meta)